AQA

FUNCTIONAL SKILLS CERTIFICATE
Functional Mathematics

Level 2

Mark Scheme

4368

November 2017

Version: 1.0 Final

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

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Glossary for Mark Schemes

Examinations are marked to award positive achievement.

Marks are awarded for demonstrating the following interrelated **process skills**.

Representing Selecting the mathematics and information to model a situation.

R.1 Candidates recognise that a situation has aspects that can be represented using mathematics.

R.2 Candidates make an initial model of a situation using suitable forms of representation.

R.3 Candidates decide on the methods, operations and tools, including ICT, to use in a situation.

R.4 Candidates select the mathematical information to use.

Analysing Processing and using mathematics.

A.1 Candidates use appropriate mathematical procedures.

A.2 Candidates examine patterns and relationships.

A.3 Candidates change values and assumptions or adjust relationships to see the effects on answers in models.

A.4 Candidates find results and solutions.

Interpreting Interpreting and communicating the results of the analysis.

I.1 Candidates interpret results and solutions.

I.2 Candidates draw conclusions in light of situations.

I.3 Candidates consider the appropriateness and accuracy of results and conclusions.

I.4 Candidates choose appropriate language and forms of presentation to communicate results and solutions.

In particular, individual marks are mapped onto the following **skills standards**.

Representing Making sense of the situations and representing them.
A learner can:

Ra Understand routine and non-routine problems in familiar and unfamiliar contexts and situations.

Rb Identify the situation or problems and identify the mathematical methods needed to solve them.

Rc Choose from a range of mathematics to find solutions.

Analysing Processing and using the mathematics.
A learner can:

Aa Apply a range of mathematics to find solutions.

Ab Use appropriate checking procedures and evaluate their effectiveness at each stage.

Interpreting Interpreting and communicating the results of the analysis.
A learner can:

Ia Interpret and communicate solutions to multistage practical problems in familiar and unfamiliar contexts and situations.

Ib Draw conclusions and provide mathematical justifications.

To facilitate marking, the following categories are used:

M Method marks are awarded for a correct method which could lead to a correct answer.

A Accuracy marks are awarded when following on from a correct method. It is not necessary to always see the method. This can be implied.

B Marks awarded independent of method.

ft Follow through marks. Marks awarded following a mistake in an earlier step.

SC Special case. Marks awarded within the scheme for a common misinterpretation which has some mathematical worth.

oe Or equivalent. Accept answers that are equivalent.
eg, accept 0.5 as well as $\frac{1}{2}$

Q	Answer	Mark	Comments
1(a)	2.21 (pm) or 14 21	B2 *Aa* *Aa*	B1 2.01(pm) or 14 01 seen or B1ft correct time for their 2.01(pm) SC1 2.41 (pm) or 1421 or 3.01 (pm) or 1501
	Additional Guidance		
	Allow any representation of correct times. E.g. 2:21 or 14.21 or 21 minutes past 2		

<table>
<tr><th>Q</th><th colspan="3">Answer</th><th>Mark</th><th colspan="3">Comments</th></tr>
<tr><td rowspan="5">1(b)</td><td colspan="3">Alternative method 1</td><td></td><td colspan="3"></td></tr>
<tr><td colspan="3">56 + 43 + 39 + (56 – 39) + (43 – 18) + 18
or
56 + 43 + 39 + 17 + 25 + 18
or
(43 + 56) × 2
or
198</td><td>M1
Aa</td><td colspan="3">allow up to two errors or omissions or one additional length</td></tr>
<tr><td colspan="3">their 198 × 4 or 792</td><td>M1
Ra</td><td colspan="3">or their 198 ÷ 150 or 1.32
their 198 must be a length</td></tr>
<tr><td>their 792 ÷ 150</td><td>5 × 150 or 750</td><td>M1
Rc</td><td>their 1.32 × 4</td><td>their 792 ÷ 5</td><td>their 750 ÷ 4</td></tr>
<tr><td colspan="3">5.2(8 …) or 5.3 or 6 and No
or
792 and 750 and No
or
158.4 and No
or
187.5 and No</td><td>A2
Ib
Ib</td><td colspan="3">A1 5.2(8 …) or 5.3
or
792 and 750
or
158(.4)
or
187.5
or
A1ft correct decision for their value(s) for A1ft must score M1M1M1 or M1M0M1 or M0M1M1 - their 198 must be a length</td></tr>
</table>

Q	Answer	Mark	Comments
1(b)	**Alternative method 2**		
	56 + 43 + 39 + (56 – 39) + (43 – 18) + 18 or 56 + 43 + 39 + 17 + 25 + 18 or (43 + 56) × 2 or 198	M1 *Aa*	allow up to two errors or omissions or one additional length
	150 × 5 or 750	M1 *Ra*	
	their 750 ÷ their 198	M1 *Rc*	their 198 must be a length
	3.78 and No	A2 *Ib* *Ib*	A1 3.78 or A1ft correct decision for their value(s) must score M1M1M1 or M0M1M1 or M1M0M1
	Additional Guidance		
	Award first M1 for any evidence of trying to find the total perimeter. '(He will need) 6 rolls' implies 'No'		
	750 – 792 and need another 42 m	M3A2	
	150 – 39 – 25 – 17 – 18 – 56 – 43 or -48 4 × their -48 or -192 150 – 192 or -42 42 m short with 5 rolls so needs 6 rolls	M1 M1 M1 A2	
	If their 198 < 198 can score M3 A1ft (allow 4 × 150 instead of 5 × 150). E.g.		
	39 + 18 + 56 + 43 = 156 4 × 156 = 624 4 × 150 = 600 Yes	M1 M1 M1 A1ft	

1(c)	**Alternative method 1**

Q	Answer	Mark	Comments
	any appropriate area calculated, eg	M1	
	56 × 43 or 2408 or (56 – 39) × (43 – 18) or 17 × 25 or 425 39 × 43 or 1677 or 18 × (56 – 39) or 18 × 17 or 306 18 × 56 or 1008 or (43 – 18) × 39 or 25 × 39 or 975	*Ra*	
	their 2408 – their 425 or their 1677 + their 306 or their 1008 + their 975 or 1983	M1 *Aa*	
	their 1983 × 1.196 or [2371, 2371.7]	M1 *Rb*	2371.668 their 1983 must be an area
	4840 ÷ 2 or 2420 / their [2371, 2372] × 2	M1 *Rc*	or their [2371, 2372] ÷ 4840
	[2371, 2372] and 2420 and No or [4742, 4744] and No	A2 *Ib* *Ib*	A1 [2371, 2372] and 2420 or [4742, 4744] or 0.48(9 …) or 0.49 or A1ft correct decision for their value(s) must score M3 and use 1.196

1(c)	**Alternative method 2**		
	any appropriate area calculated, eg	M1	

Q	Answer		Mark	Comments
	56 × 43 or 2408 or (56 – 39) × (43 – 18) or 17 × 25 or 425 39 × 43 or 1677 or 18 × (56 – 39) or 18 × 17 or 306 18 × 56 or 1008 or (43 – 18) × 39 or 25 × 39 or 975		*Ba*	
	their 2408 – their 425 or their 1677 + their 306 or their 1008 + their 975 or 1983		M1 *Aa*	
	4840 ÷ 2 or 2420	4840 ÷ 1.196 or 4046.8 …	M1 *Rb*	
	their 2420 ÷ 1.196	their 1983 × 2 or their 4046.8 ÷ 2	M1 *Rc*	2023.411 … or their 1983 ÷ their 4046.8 …
	2023.4 … and 1983 and No or 4046.8 … and 3966 and No		A2 *Ib* *Ib*	A1 2023.4 … and 1983 or 4046.8 … and 3966 or 0.48(9 …) or 0.49 or A1ft correct decision for their value(s) must score M3 and use 1.196

	Additional Guidance
1(c)	Use of perimeter instead of area can score M1 max for 4840 ÷ 2 or 2420 Error in calculation of area can score M1M0M1M1A1ft max

Q	Answer	Mark	Comments
	Misread of 1.196 (e.g. 1.96) can score M4A1ft max Not using 1.196 or misread of 1.196 can score M1M1M0M1A0 max Examples		
	Area = 1983 1983 ÷ 1.196 = 1658 2420 Compares 1658 and 2420 and states or implies No Scores A1ft as M3 scored and 1.196 used	M2 M0 M1 A1ft	
	Area = 2408 2408 ÷ 1.196 = 2013.4 2420 Compares 2013.4 and 2420 and states or implies No Does **not** score A1ft as only M2 scored even though 1.196 used	M1 M0 M1 A0ft	
	Area = 2408 2408 × 1.96 = 4719.68 2420 Compares 4719.68 and 2420 and states or implies Yes Would have scored M4A1ft with correct area	M1 M1 (mr) M1 A1ft	
	Area = 1983 4840 × 1.196 = 5788.64 5788.64 ÷ 2 = 2894.32 (5788.64 is their 4046.8 …) Compares 1983 with 2894.32 and states or implies No Scores A1ft as M3 scored and 1.196 used (even though used incorrectly)	M2 M0 M1 A1ft	

2(a)	rota completed with criteria for workers required correct	B1 *Ia*	
	rota completed with criteria for when workers are available correct	B1 *Ia*	maximum of 2 shifts per worker per day

Q	Answer	Mark	Comments
	rota clearly communicated for all shifts for all days and all names	B1 *Ia*	SC1 Workers 'correctly' allocated with shifts not specified

Additional Guidance

B1B1B1 responses

Example 1 (tabulated list)

Monday	Evening	Amy	
Tuesday	Evening	Amy	
Wednesday	Evening	Amy	
Thursday	Morning	Cathy	Ben
Thursday	Afternoon	Cathy	Ben
Thursday	Evening	Amy	Eva
Friday	Morning	Ben	Eva
Friday	Afternoon	Ben	David
Friday	Evening	David	Amy
Saturday	Morning	David	Cathy
Saturday	Afternoon	Ben	David
Saturday	Evening	Ben	Amy
Sunday	Afternoon	Cathy	

Example 2 (non-tabulated list)

Amy – Mon, Tue, Wed, Thu, Fri and Sat Evenings
Ben – Thu Morning and Afternoon, Fri Morning and Afternoon, Sat Afternoon and Evening
Cathy – Thu Morning and Afternoon, Sat Morning, Sun Afternoon
David – Fri Afternoon and Evening, Sat Morning and Afternoon
Eva – Thu Evening and Fri Morning

2(a)

Example 3 (2-way table)

	Mon	**Tue**	**Wed**	**Thu**	**Fri**	**Sat**	**Sun**
Amy	Evening	Evening	Evening	Evening	Evening	Evening	
Ben				Morning Afternoon	Morning Afternoon	Afternoon Evening	

Q	Answer	Mark	Comments

Cathy				Morning Afternoon		Morning	Afternoon	
David					Afternoon Evening	Morning Afternoon		
Eva				Evening	Morning			

Example 4 (2-way table)

	Mon	**Tue**	**Wed**	**Thu**	**Fri**	**Sat**	**Sun**
Morning				B and C	B and E	C and D	
Afternoon				B and C	B and D	B and D	C
Evening	A	A	A	A and E	A and D	A and B	

For B1B1B1 there must be a total of 22 "worker allocations"

Same worker allocated to the same shift more than once could score B0B0B1 max

Same worker with more than 2 shifts a day could score B1B0B1 max

Specifying shifts for Thu, Fri and Sat only could score B1B1B0

No shifts specified - final mark for communication must be B0

The final communication mark can be scored for

a fully labelled 2-way table completed incorrectly

an incorrect list including all names, all shifts and all days

2(b)	**Alternative method 1**			
	1800 – 6 × 20 or 1800 – 120 or 1680	(1800 – 20) ÷ 5 or 356	M1 *Rb*	or dimensions correctly converted to cm

Q	Answer	Mark	Comments
	their 1680 ÷ 5 their 356 – 20	M1 *Aa*	their 1680 cannot be 1800
	336 and Yes	A2 *Ib* *Ib*	A1 336 or A1ft correct decision from their value must score M2 or M1M0 or M0M1 with 5 shelves instead of 6 or 6 spaces instead of 5
	Alternative method 2		
	320 × 5 or 1600 or 6 × 20 or 120	M1 *Rb*	or dimensions correctly converted to cm
	their 1600 + their 120	M1 *Aa*	
	1720 and Yes	A2 *Ib* *Ib*	A1 1720 or A1ft correct decision from their value must score M2 or M1M0 or M0M1 with 5 shelves instead of 6 or 6 spaces instead of 5

2(b)	**Alternative method 3**		
	1800 – 6 × 20 or 1800 – 120 or 1680	M1 *Rb*	or dimensions correctly converted to cm
	320 × 5	M1 *Aa*	1680 ÷ 320

Q	Answer	Mark	Comments
	1680 and 1600 and Yes	A2	A1 1680 and 1600
	or 5.25 and Yes	*Ib* *Ib*	or 5.25 or A1ft correct decision from their value must score M2 or M1M0 or M0M1 with 5 shelves instead of 6 or 6 spaces instead of 5
	Additional Guidance		
	Starting with 1800 ÷ 6 scores zero Mixing up 5 and 6 can occur once only for A1ft Examples that score M1M0A1ft or M0M1A1ft		
	Alt 1 1680 1680 ÷ 6 = 280 No		 M1 M0 A1ft
	Alt 2 320 × 6 = 1920 and 5 × 20 = 100 1920 + 100 = 2020 Yes Uses 5 shelves and 6 spaces so mixes up 5 and 6 twice		 M0 M1 A0
	Alt 3 1800 – 5 × 20 = 1700 320 × 5 = 1600 Yes		 M0 M1 A1ft

2(c)	550 ÷ 260 or 2.1(1…) or 2.12 or 2 or 470 ÷ 155 or 3.0(3…) or 3 or 390 ÷ 65 or 6	M1 *Ra*	Allow 155 × 3 = 465 etc or 470 – 155 – 155 – 155 (= 5) etc
	their 2 and their 3 and their 6	M1 *Ib*	their values rounded down award M2 if first M1 scored and 11 seen

Q	Answer	Mark	Comments
	their 2 × their 3 × their 6	M1 *Rc*	multiplying their values allow if incorrectly rounded or not rounded
	36	A1 *Aa*	
	Additional Guidance		
	These attempts are common and score zero • (470 × 390 × 550) ÷ (155 × 65 × 260) or 38 • (470 + 390 + 550) ÷ (155 + 65 + 260) or 3		

Q	Answer		Mark	Comments
2(d)	**Alternative method 1**			
	11.41 ÷ 1000 or (£) 0.01141 or 1.141 p		M1 *Ra*	allow subsequent use of 0.01 or 1 if method seen
	their 0.01141 × 384 or 4.38144 or 0.05 × 384 or 19.2	their 1.141 × 384 or 438.144 or 5 × 384 or 1920	M1 *Rc*	
	their 19.2 – their 4.38144 or 14.81856	their 1920 – their 438.144 or 1481.856	M1 *Aa*	units must be consistent
	their 14.81856 × 0.8	their 1481.856 × 0.8	M1 *Aa*	
	£11.85 or £11.86 or 1185 p or 1186 p		A1 *Ia*	must see £ or p
	Alternative method 2			
	11.41 ÷ 1000 or (£) 0.01141 or 1.141 p		M1 *Ra*	allow subsequent use of 0.01 or 1 if method seen
	0.05 – 0.01141 or 0.03859	5 – 1.141 or 3.859	M1 *Rc*	profit on one bag in £ or p units must be consistent
	384 × their 0.03859 or [14.81, 14.82]	384 × their 3.859 or [1481, 1482]	M1 *Aa*	profit on 384 bags in £ or p
	their [14.81, 14.82] × 0.8	their [1481, 1482] × 0.8	M1 *Aa*	
	£11.85 or £11.86 or 1185 p or 1186 p		A1 *Ia*	must see £ or p

Q	Answer		Mark	Comments
2(d)	**Alternative method 3**			
	11.41 ÷ 1000 or (£) 0.01141 or 1.141 p		M1 *Ra*	allow subsequent use of 0.01 or 1 if method seen
	their 0.01141 × 384 or 4.38144 or 0.05 × 384 or 19.2	their 1.141 × 384 or 438.144 or 5 × 384 or 1920	M1 *Rc*	
	their 19.2 × 0.8 or 15.36 or their 4.38144 × 0.8 or 3.505 …	their 1920 × 0.8 or 1536 or their 438.144 × 0.8 or 350.5	M1 *Aa*	units must be consistent
	their 15.36 – their 3.505	their 1536 – their 350.5	M1 *Aa*	
	£11.85 or £11.86 or 1185 p or 1186 p		A1 *Ia*	must see £ or p
	Alternative method 4			
	0.05 × 1000 or (£)50	5 × 1000 or 5 000 p	M1 *Ra*	
	their 50 – 11.41 or (£)38.59	their 5 000 – 1141 or 3 859 p	M1 *Rc*	profit on 1000 bags in £ or p units must be consistent
	384 ÷ 1000 × their 38.59 or [14.81, 14.82]	384 ÷ 1000 × their 3859 or [1481, 1482]	M1 *Aa*	
	their 0.8 × their [14.81, 14.82]	their 0.8 × their [14.81, 14.82]	M1 *Aa*	
	£11.85 or £11.86 or 1185 p or 1186 p		A1 *Ia*	must see £ or p

Q	Answer		Mark	Comments
2(d)	**Alternative method 5**			
	11.41 ÷ 1000 or (£) 0.01141 or 1.141 p		M1 *Ra*	allow subsequent use of 0.01 or 1 if method seen
	0.05 – 0.01141 or 0.03859	5 – 1.141 or 3.859	M1 *Rc*	profit on one bag in £ or p units must be consistent
	their 0.03859 × 0.8 or 0.030872	their 3.859 × 0.8 or 3.0872	M1 *Aa*	80% of profit on one bag in £ or p
	their 0.030872 × 384	their 3.0872 × 384	M1 *Aa*	
	£11.85 or £11.86 or 1185 p or 1186 p		A1 *Ia*	must see £ or p
	Alternative method 6			
	11.41 ÷ 1000 or (£) 0.01141 or 1.141 p		M1 *Ra*	allow subsequent use of 0.01 or 1 if method seen
	their 0.01141 × 0.8 or 0.009128 or 0.05 × 0.8 or 0.04	their 1.141 × 0.8 or 0.9128 or 5 × 0.8 or 4	M1 *Rc*	
	their 0.009128 × 384 or 3.5051 … or their 0.04 × 384 or 15.36	their 0.9128 × 384 or 350 51 … or their 4 × 384 or 1536	M1 *Aa*	
	their 15.36 – their 3.5051 …	their 1536 – their 3505.1	M1 *Aa*	
	£11.85 or £11.86 or 1185 p or 1186 p		A1 *Ia*	must see £ or p

<table>
<tr><th>Q</th><th>Answer</th><th>Mark</th><th colspan="2">Comments</th></tr>
<tr><td rowspan="6">2(d)</td><td colspan="4">Additional Guidance</td></tr>
<tr><td colspan="4">Incorrect answer caused by premature or incorrect rounding or truncation can score M4A0
Award M1 for the correct method for working out 80% of any sum of money or 80% of 384</td></tr>
<tr><td colspan="4">There are other less obvious methods. E.g.
Working out the profit as a fraction of the cost multiplied by 80% of the cost from
[(50 – 11.41) ÷ 50] × 19.2 × 0.8</td></tr>
<tr><td colspan="4">2 marks can be scored for <u>only</u> working out the total sales of 384 bags (£15.36)
3 marks can be scored for <u>only</u> working out the total sales (£15.36) plus the unit cost per bag (£0.01141)
3 marks can be scored for only working out the total cost of 384 bags (£[3.50.3.51])
Note that £15.36 can also be obtained from the profit per bag if the unit cost per bag has been rounded to 1p (giving a profit per bag of 4p)
Examples</td></tr>
<tr><td>11.41 ÷ 1000 = 0.01
0.05 – 0.01 = 0.04
0.04 × 384 = 15.36
0.8 × 15.36 = £12.29</td><td>M1
M1
M1
M1A0</td><td>0.01 (rounded down - no method)
0.05 – 0.01 = 0.04
0.04 × 384 = 15.36
0.8 × 15.36 = £12.29</td><td>M0
M1
M1
M1A0</td></tr>
</table>

Q	Answer	Mark	Comments
3(a)	225 litres	B1 *Aa*	

3(b)	3 × 77 or 231		M1 *Ra*	water used for 3 baths
	216 ÷ 6 × 3 or 108		M1 *Rc*	water used for 3 showers 123 implies M2
	1.5 × 30 or 45		M1 *Ra*	water saved for 30 flushes or (288 ÷ 32) × 30 – ((288 ÷ 32) – 1.5) × 30 or their 9 × 30 – their 7.5 × 30 or their 270 – their 225
	their 231 – their 108 + their 45 or their 123 + their 45 or (270 + 231) – (225 + 108) or 501 – 333 or 168		M1 *Rc*	total water saved
	1200 ÷ 6	their 168 × 6	M1 *Aa*	1 ÷ 6 = 0.16(6 …) and their 168 ÷ 1200 — 1200 ÷ their 168
	168 and 200 and No or 1008 and No or 0.16(6 …) or 0.17 and 0.14 and No or 7.14 … and No		A2 *Ib* *Ib*	A1 168 and 200 or 1008 or 0.16(6 …) or 0.17 and 0.14 or 7.14 or A1ft correct conclusion for their value(s) must score M4 including 4th and 5th M1

Q	Answer	Mark	Comments
	Additional Guidance		
3(b)	1032 (from 1200 – 168) implies M4 and could score full marks, e.g. 1032 compared with 1000 $^{168}/_{1200}$ simplified to $^{7}/_{50}$ with correct conclusion scores full marks with no further work misread of 288 (e.g. 228) This should still give 45 litre saving for toilet flushes but there could be rounding errors, e.g. 228 ÷ 32 = 7.13 (should be 7.125) with misread (with or without rounding errors) can score M5A1 or M5 A1ft		

Q	Answer	Mark	Comments
3(c)	450 × their 365 ÷ 1000 or 0.45 × their 365 or their 164 250 ÷ 1000	M1 *Ra*	their 365 can be 364, 366 or 360 364 → 163.8 366 → 164.7 360 → 162
	164.25	A1 *Aa*	allow 164 if with method
	an appropriate reverse method, e.g; 164.25 × 1000 ÷ their 365 = 450	B1ft *Ab*	allow either of these following estimations 500 × 400 ÷ 1000 = 200 or 400 × 400 ÷ 1000 = 160
	Additional Guidance		
	163.8 or 164.7 or 162 scores M1A0 Mark holistically, e.g. if there is an incorrect answer with no method in the main body mark the method and/or answer in the check (and vice versa) The correct reverse method is only a valid check if the original method is shown Using approximations can score the checking mark without any other method All values used to calculate their 164.25 must be used for a reverse method to score the checking mark If the original method is not shown fully but the correct answer is obtained, do not award the checking mark for a reverse method unless it is complete e.g 450 × 365 = 164 250 litres = 164.25 m^3 (division by 1000 not shown) followed by 164 250 ÷ 450 = 365 scores M1A1B0 450 × 365 = 164 250 litres = 164.25 m^3 (division by 1000 not shown) followed by 164.25 × 1000 ÷ 450 = 365 scores M1A1B1		

Q	Answer	Mark	Comments
3(d)	~~2.96 × their 164.25 or 486.18~~ or 2.96 × their 164 or 485.44	M1 *Rc*	~~ft their 164(.25) from 3(c)~~
	their 486.18 + 134 or 620.18 or their 485.44 + 134 or 619.44	M1 *Aa*	
	12 × 53.5(0) or 642	M1 *Aa*	
	620.18 and 642 or 21.82 and Yes or 619.44 and 642 or 22.56 and Yes	A2ft *Ib* *Ib*	A1ft 620.18 and 642 or 21.82 or 619.44 and 642 or 22.56 or A1ft correct conclusion for their values must score M3 ft their 164(.25) from 3(c)
	Additional Guidance		
	'With a water meter is cheaper' implies Yes ft from 3(c) – allow rounding or truncation of their value Common examples scoring full follow through marks		
	If their 164(.25) = 163.8 2.96 × 163.8 = 484.85 484.85 + 134 = 618.85 12 × 53.5 = 642 Yes	If their 164(.25) = 164.7 2.96 × 164.7 = 487.51 487.51 + 134 = 621.51 12 × 53.5 = 642 Yes	If their 164(.25) = 162 2.96 × 162 = 479.52 479.52 + 134 = 613.52 12 × 53.5 = 642 Yes

Q	Answer	Mark	Comments
4(a)	**Alternative method 1**		
	the fastest 8 swimmers chosen	B2 *Ra* *Rb*	B1 7 of the fastest 8 swimmers chosen
	all of their 8 in correct lanes	B2ft *Rc* *Aa*	B1ft the fastest of their 8 in lane 4 or Jack in lane 4
	Alternative method 2 (only times given)		
	B3 the 8 fastest times in 'correct' lanes	B3 *Ra* *Rb* *Rc*	B2 the 8 fastest times in time order or B1 the 8 fastest times given

<table>
<tr><th>Q</th><th>Answer</th><th>Mark</th><th>Comments</th></tr>
<tr><td rowspan="3">4(a)</td><td colspan="3">Additional Guidance</td></tr>
<tr><td colspan="3">mark the table only</td></tr>
<tr><td colspan="3">
<table>
<tr><th>Name</th><th>Time</th><th>Lane in final</th></tr>
<tr><td>Jack</td><td>52.83</td><td>4</td></tr>
<tr><td>Kev</td><td>52.88</td><td>5</td></tr>
<tr><td>Ahmed</td><td>52.92</td><td>3</td></tr>
<tr><td>Paul</td><td>52.93</td><td>6</td></tr>
<tr><td>Cheng</td><td>52.97</td><td>2</td></tr>
<tr><td>Zain</td><td>53.14</td><td>7</td></tr>
<tr><td>Dai</td><td>53.20</td><td>1</td></tr>
<tr><td>Tom</td><td>53.23</td><td>8</td></tr>
<tr><td>Ian</td><td>53.26</td><td></td></tr>
<tr><td>Mike</td><td>53.28</td><td></td></tr>
<tr><td>Yan</td><td>53.28</td><td></td></tr>
<tr><td>Ralf</td><td>53.37</td><td></td></tr>
<tr><td>Simon</td><td>53.49</td><td></td></tr>
<tr><td>Louis</td><td>53.57</td><td></td></tr>
<tr><td>Greg</td><td>53.66</td><td></td></tr>
<tr><td>Harry</td><td>53.70</td><td></td></tr>
</table>
<table>
<tr><td>Lane 1</td><td>Dai</td><td>53.20</td></tr>
<tr><td>Lane 2</td><td>Cheng</td><td>52.97</td></tr>
<tr><td>Lane 3</td><td>Ahmed</td><td>52.92</td></tr>
<tr><td>Lane 4</td><td>Jack</td><td>52.83</td></tr>
<tr><td>Lane 5</td><td>Kev</td><td>52.88</td></tr>
<tr><td>Lane 6</td><td>Paul</td><td>52.93</td></tr>
<tr><td>Lane 7</td><td>Zain</td><td>53.14</td></tr>
<tr><td>Lane 8</td><td>Tom</td><td>53.23</td></tr>
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</table>

Q	Answer	Mark	Comments
4(b)	0.37 secs	B1 *Aa*	
	an appropriate reverse method, e.g. 51.91 + 0.37 = 52.28 or 52.28 – 0.37 = 51.91	B1ft *Ab*	or this estimation 51.9 – 52.3 = 0.4 or alternative methods, e.g. 52.28 – 51.91 = 0.37 and 51.91→ + 0.09→52→ + 0.28→52.28 and 0.09 + 0.28 = 0.37
	Additional Guidance		
	must see method for 0.37 to score checking mark using reverse or alternative method an alternative correct answer is 370 milliseconds		

Q	Answer	Mark	Comments
4(c)	**Alternative method 1**		
	Beth 65.7 + 66.6 + 67.4 + 63.6 + 65.2 + 64.8 + 66.5 + 64.9 + 68.5 + 66.8 or 660 or **Daisy** 62.5 + 63.2 + 67.4 + 62.6 + 64.2 + 66.2 + 64.8 + 65.9 or 516.8	M1 *Ra*	
	their 660 ÷ 10 or 66 or their 516.8 ÷ 8 or 64.6	M1 *Aa*	
	66 and 64.6 and Daisy	A2 *Ib* *Ib*	A1 66 and 64.6 or A1ft correct conclusion for their values must score M2
	Alternative method 2		
	Beth 63.6, 64.8, 64.9, 65.2, 65.7, 66.5, 66.6, 66.8, 67.4, 68.5 or **Daisy** 62.5, 62.6, 63.2, 64.2, 64.8 ,65.9, 66.2, 67.4	M1 *Ra*	Arranges in order and indicates middle
	(65.7 + 66.5) ÷ 2 or 66.1 or (64.2 + 64.8) ÷ 2 or 64.5	M1 *Aa*	
	66.1 and 64.5 and Daisy	A2 *Ib* *Ib*	A1 66.1 and 64.5 or A1ft correct conclusion for their values must score M2

Q	Answer	Mark	Comments
4(c)	**Alternative method 3**		
	compare proportion of times under 65 seconds Beth $^{3}/_{10}$ or Daisy $^{5}/_{8}$	M1 *Ra*	can also compare under 66 seconds ($^{5}/_{10}$ and $^{6}/_{8}$)
	Beth $^{3}/_{10}$ and Daisy $^{5}/_{8}$	M1 *Aa*	
	30% and 62.5% and Daisy or $^{12}/_{40}$ and $^{25}/_{40}$ (oe) and Daisy or clear explanation that 5 out of 8 is better than 3 out of 10 and chooses Daisy	A2 *Ib* *Ib*	A1 30% and 62.5% or $^{12}/_{40}$ and $^{25}/_{40}$ (oe) or A1ft correct conclusion for their values must score M2
	Alternative method 4		
	correct comparison of at least two of fastest and/or slowest times with no average used and Daisy	B2 *Ra* *Ib*	B1 comparison of the fastest or the slowe and Daisy or correct comparison of at least two fas and Beth
	Additional Guidance		
	one correct average (mean or median) scores M2A1 correct method with incorrect means can score M2A1ft 'Daisy has better time' implies Daisy should be chosen		

Q	Answer	Mark	Comments
4(d)	**Alternative method 1**		
	(46.95 + 6.50 + 62.95) × 6 or 116.4 × 6 or 698.4	M1 *Ra*	(46.95 + 6.50 + 62.95) ÷ 100 × 15 or 116.4 ÷ 100 × 15 or 17.46 or 1 – 0.15 or 0.85
	their 698.4 ÷ 100 × 15 or 104.76 or 1 – 0.15 or 0.85	M1 *Rc*	their 116.4 – 17.46 or their 116.4 × 0.85
	their 698.4 – their 104.76 or their 698.4 × 0.85 or 593.64	M1 *Rc*	or their 98.94 × 6 or 593.64
	their 593.64 + 370 = their 963.64 or 1000 – 370 = 630 and their 593.64	M1 *Aa*	their 593.64 must be their 85% of their 698.4 cannot be their 104.76
	593.64 + 370 = 963.64 or 1000 – 370 = 630 and 593.64	A1 *Ib*	allow rounding or truncating of their 963.64 or their 593.64 to nearest £ or 10 p if method shown

Q	Answer	Mark	Comments
4(d)	**Alternative method 2**		
	46.95 × 6 + 6.50 × 6 + 62.95 × 6 or 281.7 + 39 + 377.7 or 698.4(0)	M1 *Ra*	or 46.95 ÷ 100 × 15 or 7.0425 or 6.50 ÷ 100 × 15 or 0.975 or 62.95 ÷ 100 × 15 or 9.4425 or 1 – 0.15 or 0.85
	their 698.40 ÷ 100 × 15 or 104.76 or 1 – 0.15 or 0.85	M1 *Rc*	or 46,95 – 7.0425 or 46.95 × 0.85 or 39.9075 or 6.5 – 0.975 or 6.5 × 0.85 or 5.525 or 62.95 – 9.4425 or 62.95 × 0.85 or 53.5075
	their 698.40 – their 104.76 or their 698.40 × 0.85 or 593.64	M1 *Rc*	or (39.9075 + 5.525 + 53.5075) × 6 or 593.64
	their 593.64 + 370 = their 963.64 or 1000 – 370 = 630 and their 593.64	M1 *Aa*	their 593.64 must be their 85% of their 698.4 cannot be their 104.76
	593.64 + 370 = 963.64 or 1000 – 370 = 630 and 593.64	A1 *Ib*	allow rounding or truncating of their 963.64 or their 593.64 to nearest £ or 10 p if method shown

Q	Answer	Mark	Comments
	Additional Guidance		
4(d)	allow 46.95 + 6.50 + 62.95 × 6 = 431.15 for first M1 to score M4A1 the full method must be shown apart from the method for deducting 15% <u>deducting 15%</u> can score M1M1 if • rounded or truncated values for 15% are given or used with no method e.g. 10% = 69.8 5% = 34.9 15% = 69.8 + 34.9 = 104.7 M1 698.4 – 104.7 = 593.7 M1 593.7 + 370 = 963.7 M1A0 can score M0M1 if • an incorrect method for finding 15% is given (not + 15) e.g. 698.4 ÷ 15 = 46.56 M0 698.4 – 46.56 = 651.84 M1 scores M0M0 if • an incorrect value for 15% is given with no method shown e.g. 15% = 46.56 M0 698.4 – 46.56 = 651.84 M0 15% not deducted can score M1M0M0M1A0 <u>adding 370</u> if 370 is added to their 698.4 before deducting 15% • do not award the mark for adding 370 • award any marks for deducting 15% e.g. 281.6 + 39 + 377.7 + 370 = 1068.4 M1 (method for 698.4 embedded) 1068.4 × 0.85 or 1068.4 – their 106.26 = 908.14 M2A0		